

What Makes Consumers Accept Higher Prices?

**Price Premium, Sephora Exclusivity,
and Vegan Claim Moderation Analysis**

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Background

“

Beauty pricing is not just about being expensive or cheap.
It is about **whether consumers can accept the price**

”



1) Why It Matters?

Price affects evaluation, but its effect depends on product appeal



2) Business Relevance

As K-beauty expands globally, value, ingredients, and brand messages matter more



3) Key Question

When do consumers accept higher beauty product prices?



Price

+



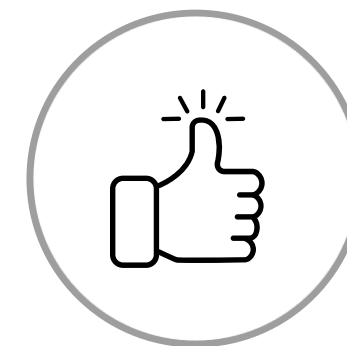
Sephora Exclusivity

+

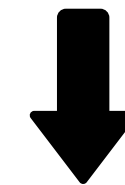


Vegan Claim

+



Rating

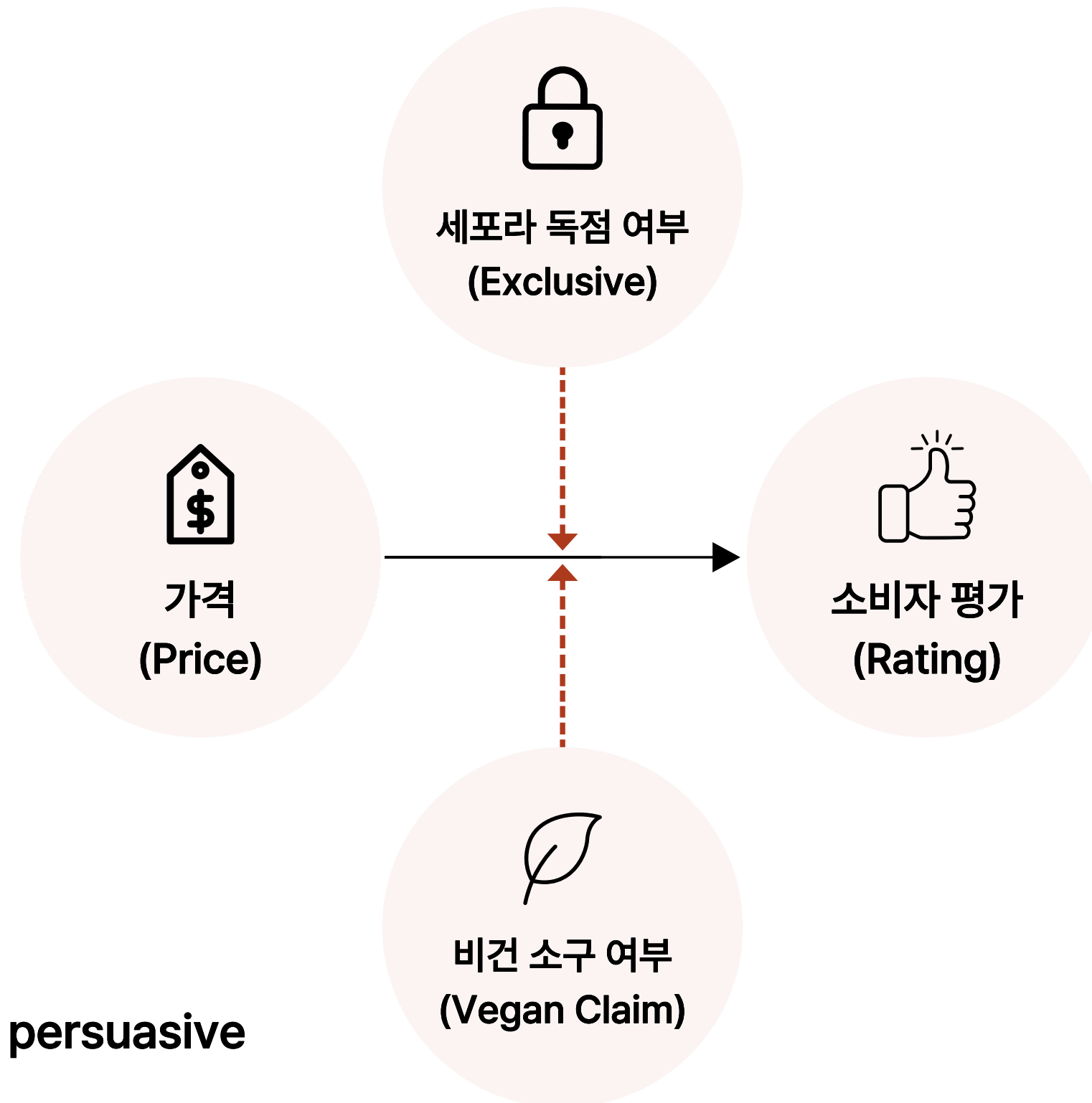


Price Acceptance

Research Questions and Objectives

“What should be considered when setting prices for higher consumer satisfaction?”

- 1 Examine the price–rating relationship
- 2 Test Sephora exclusivity as a moderator
- 3 Test vegan claims as a moderator
- 4 Identify conditions that make price premiums persuasive



Data Overview

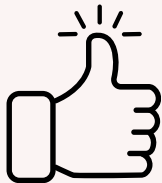
Kaggle dataset of Sephora online beauty products and user reviews, collected in March 2023. It includes about 8,000 products and 1 million reviews.

Unit	Products	Variables	Brands	Main Categories	Source
Product	8,494	67	304	9	Kaggle

Key Variables



price
가격 (USD)



rating
평점 (1~5)



reviews
리뷰 수



love_count
좋아요 수



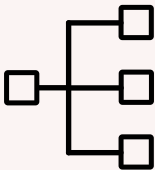
sephora_exclusive
세포라 독점 여부 (0/1)



claim_vegan
비건 여부 (0/1)

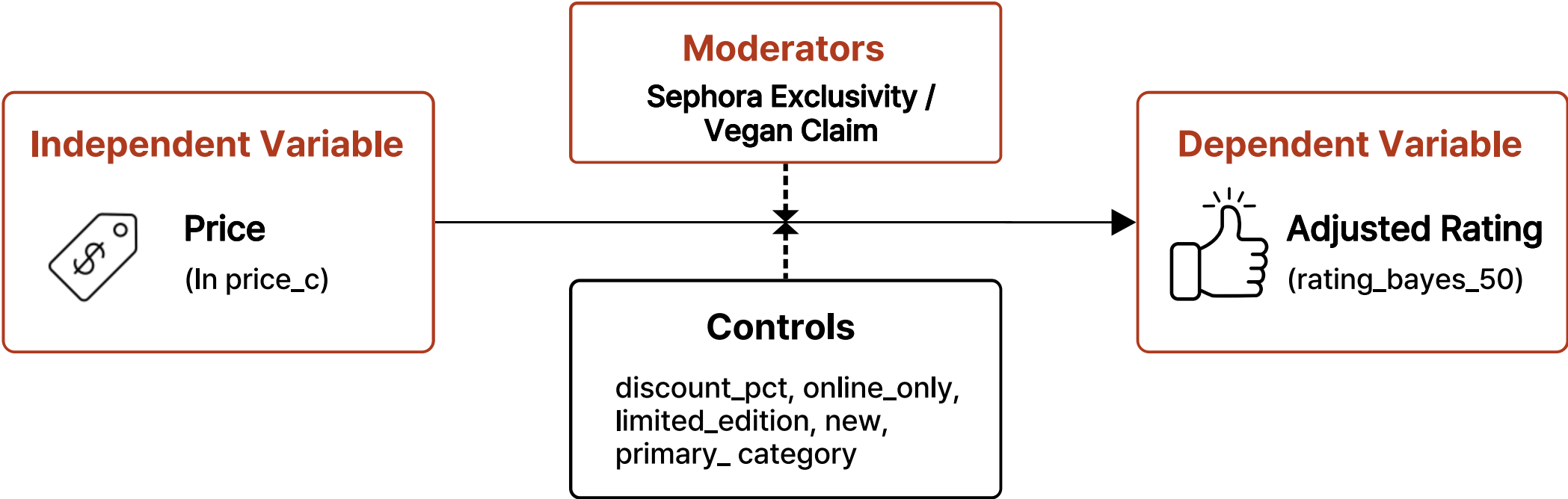


discount_pct
할인율 (%)



primary_category
대분류 카테고리

Key Variables and Conceptual Model



Role	Variable	Meaning
Independent Variable	Price (ln_price_c)	Log-transformed selling price in USD. Higher prices may signal premium value
Dependent Variable	Adjusted Rating (rating_bayes_50)	Bayesian-adjusted rating based on 50 reviews. A more stable satisfaction measure
Moderators	Sephora Exclusivity (sephora_exclusive)	Whether the product is Sephora-exclusive (1/0)
	Vegan Claim (claim_vegan)	Whether the product claims to be vegan (1/0)
Controls	discount_pct	Discount rate; controls actual price burden
	online_only	Online-only status; controls channel restriction
	limited_edition	Limited-edition status; controls scarcity effects
	new	New product status; controls newness effects
	primary_category	Main product category; controls category differences

Hypotheses

This study tests whether Sephora exclusivity and vegan claims change the price–rating relationship

H1

H0: Price and rating are not related

H1: Price and rating are related

H2

H0: **Sephora exclusivity** does not moderate the price–rating relationship

H1: **Sephora exclusivity** moderates the price–rating relationship

H3

H0: **Vegan claims** do not moderate the price–rating relationship

H1: **Vegan claims** moderate the price–rating relationship

Analysis Direction: First, test the price–rating relationship. Then, test how exclusivity and vegan claims moderate it

→ Ultimately, the focus is not just the price–satisfaction relationship, **but the conditions that make the price acceptable**

Analysis Design

This study tests how price affects ratings,
then examines whether exclusivity and vegan claims change this effect
The goal is not prediction, but **explaining what makes a price acceptable**

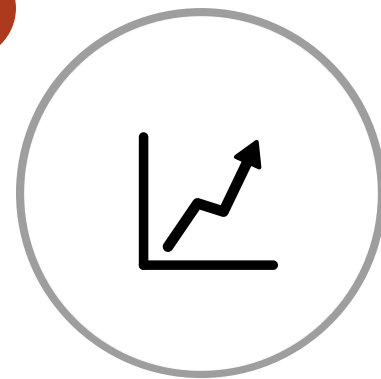
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Exploratory Analysis

Price distribution, ratings by
price range, and group
differences

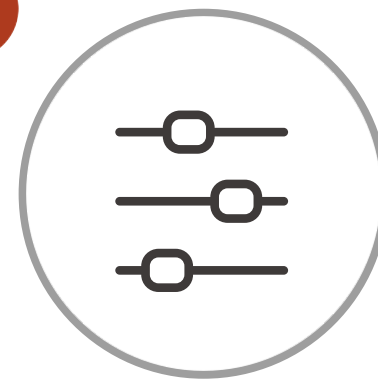
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H1 Test

Price–rating relationship
using regression

3



H2 · H3 Test

Moderation effects:
Price × Exclusivity / Price
× Vegan Claim

4



Additional Analysis

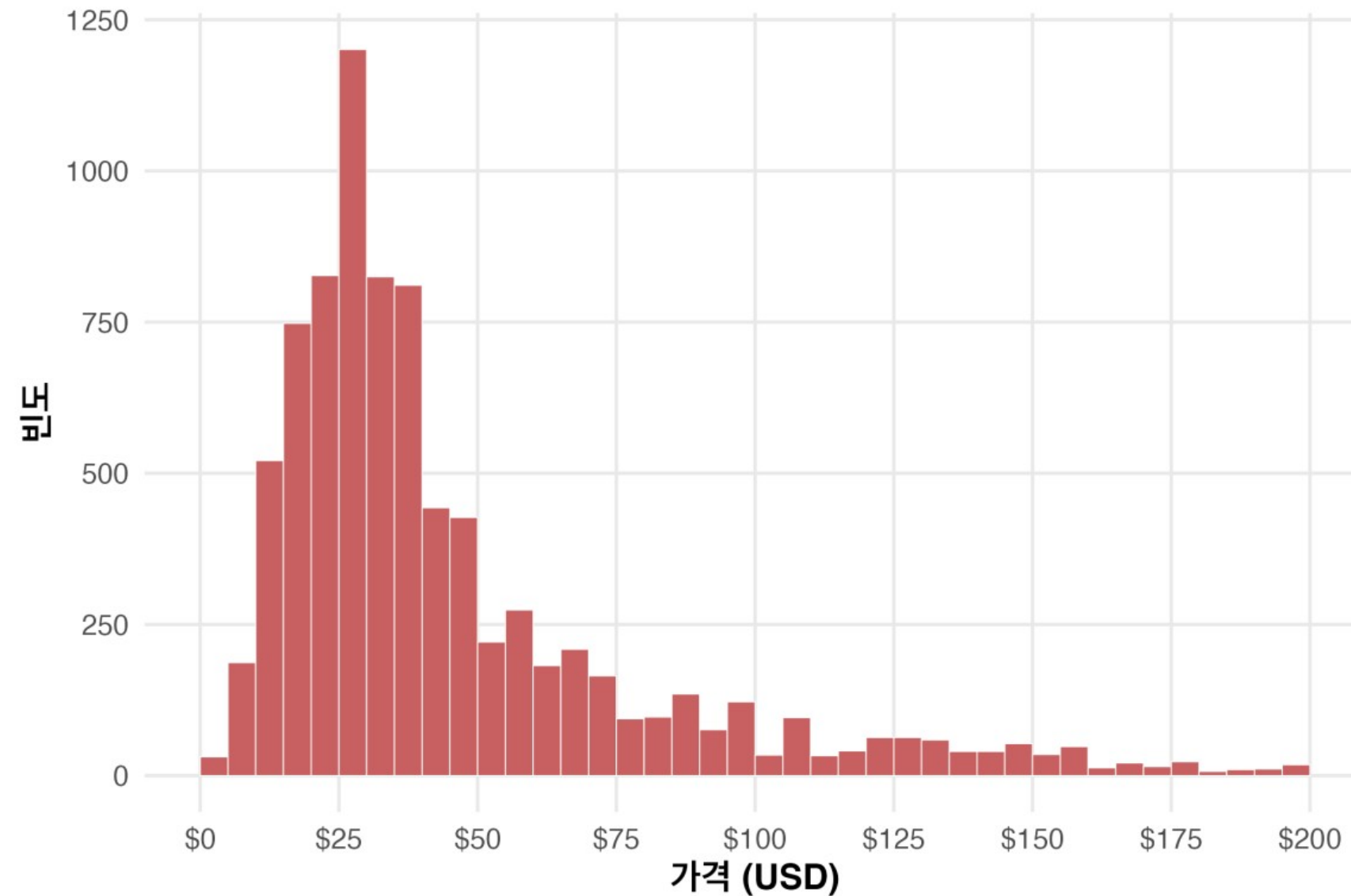
Binary logit using high_rating
and category-level analysis

Why This Method?

- Adjusted rating is continuous, so multiple regression is used
- H1 tests the baseline price effect
- H2 and H3 test moderation through interaction terms
- Binary logit checks robustness using high_rating

Exploratory Analysis 1: Price Distribution

가격 분포: 5달러 단위 구간



Insight

- Most products are concentrated in the \$20–40 range
- A small number of high-priced products form a premium segment



Prices below \$200 were visualized in 5-dollar bins because the original price range was too wide

Exploratory Analysis 2: Price–Rating Trend

가격대별 평균 보장 평점

단순 시각화에서는 약한 역U자형 가능성이 관찰됨



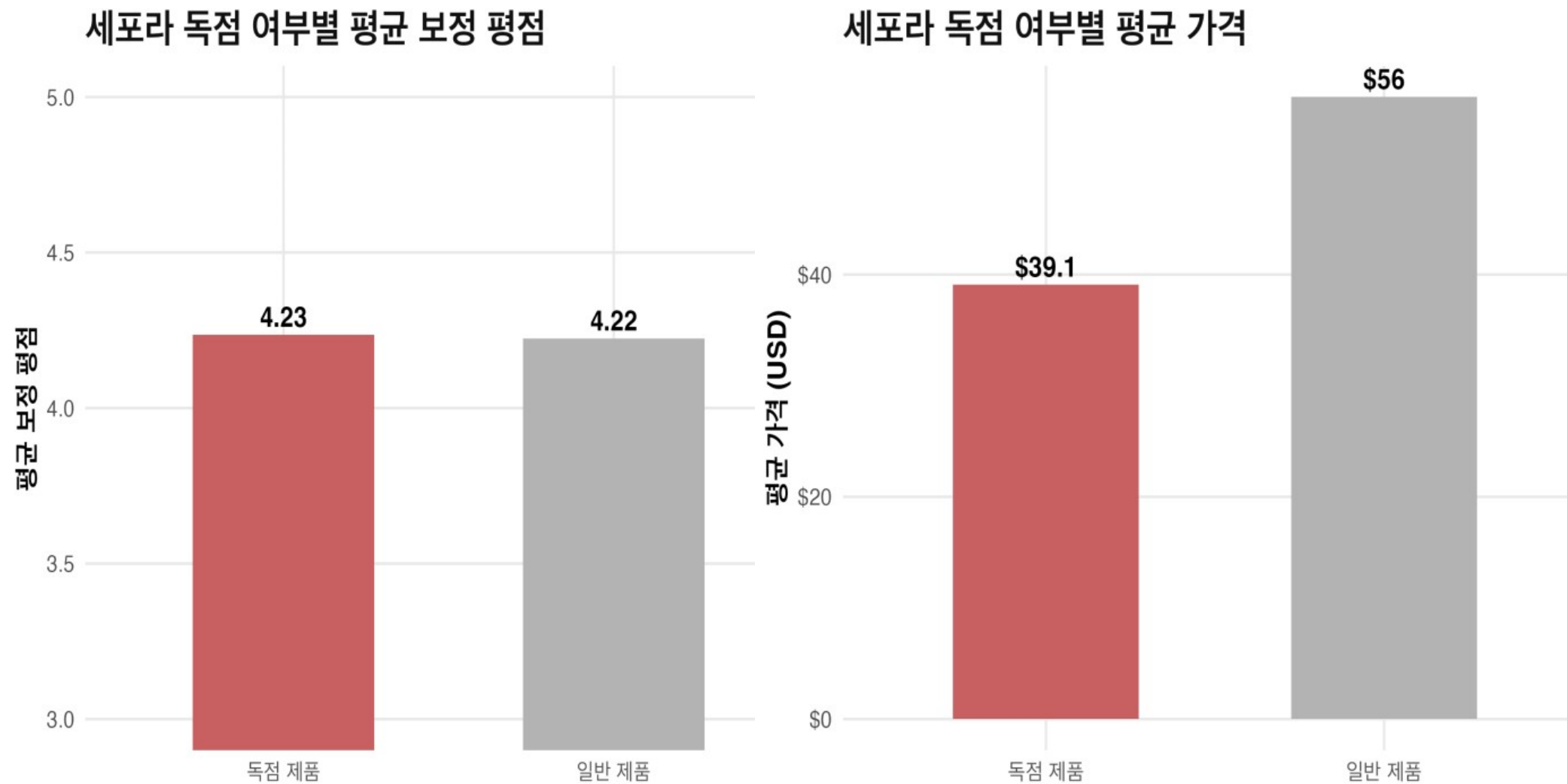
Insight

- Ratings rise with price in the low-price range
- Ratings peak around \$60–70, then decline at very high prices
- The price effect appears nonlinear, not simply linear



A weak inverted U-shape was observed, so H1 includes a quadratic price term

Exploratory Analysis 3: Sephora Exclusivity



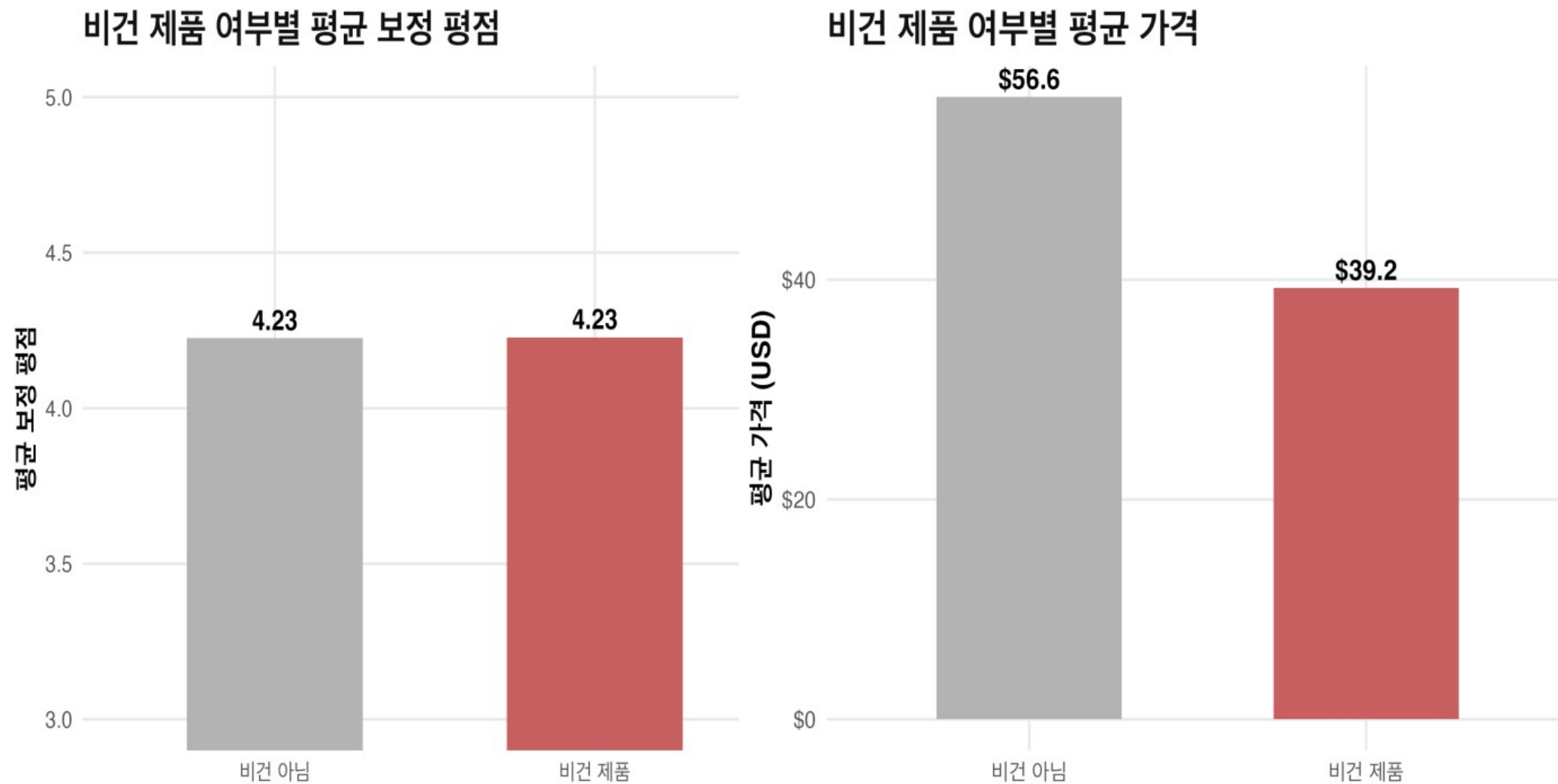
Insight

- Exclusive and non-exclusive products show similar ratings
- Exclusive products are not priced higher on average
- The key is whether exclusivity changes price acceptance



Since the simple mean comparison does not clearly show the role of exclusivity, the following analysis tests the moderating effect of Price × Exclusivity

Exploratory Analysis 4: Vegan Claim



Insight

- Vegan and non-vegan products show almost identical ratings
- Vegan products are priced lower on average
- The key is whether vegan claims change how consumers interpret price



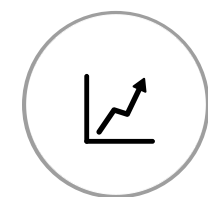
Since the simple mean effect of vegan claims is not clear, the following analysis tests the moderating effect of Price × Vegan Claim

Model Specification

Method: Least-Squares Multiple Regression

|

DV: Adjusted Rating (rating_bayes_50)



H1
Price–Rating Relationship
Baseline Model with Price²

Equation

Adjusted Rating = b0 + b1 Price + b2 Price² + Controls + e

Purpose

- Test the inverted U-shaped pattern
- Include Price² to capture nonlinearity
- Set the baseline model



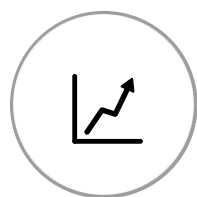
H2
Exclusivity Moderation
Does exclusivity change the price–rating relationship?

Equation

Adjusted Rating = b0 + b1 Price + b2 Exclusivity + b3 Price × Exclusivity + Controls + e

Purpose

- Test the main effect of exclusivity
- Test Price × Exclusivity interaction
- Check whether exclusivity changes price acceptance



H3
Vegan Claim Moderation
Do vegan claims change the price–rating relationship?

Equation

Adjusted Rating = b0 + b1 Price + b2 Exclusivity + b3 Price × Exclusivity + b4 Vegan + b5 Price × Vegan + Controls + e

Purpose

- Test the main effect of vegan claims
- Test Price × Vegan interaction
- Check whether vegan claims change price acceptance



Why Include These?

1. H1 was revised to include Price² after an inverted U-shaped pattern appeared
2. H2 and H3 test moderation through Price × Exclusivity and Price × Vegan interactions
3. All models include discount, online-only, limited edition, newness, and category controls

Model Analysis 1: Price–Rating Inverted U

H1 Result (DV: Adjusted Rating)

Conclusion: Significant **inverted U-shape**

Regression Results

- 1

Price **$b1 = 0.053$** ($p < 0.05$)

Significant positive effect
→ Higher price predicts higher rating
- 2

Price² **$b2 = -0.029$** ($p < 0.05$)

Significant negative effect
→ Rating growth slows after a certain point
- 3

Optimal Point $\frac{\exp(-b1 / (2 \cdot b2))}{+ \text{mean log price}} \approx \96

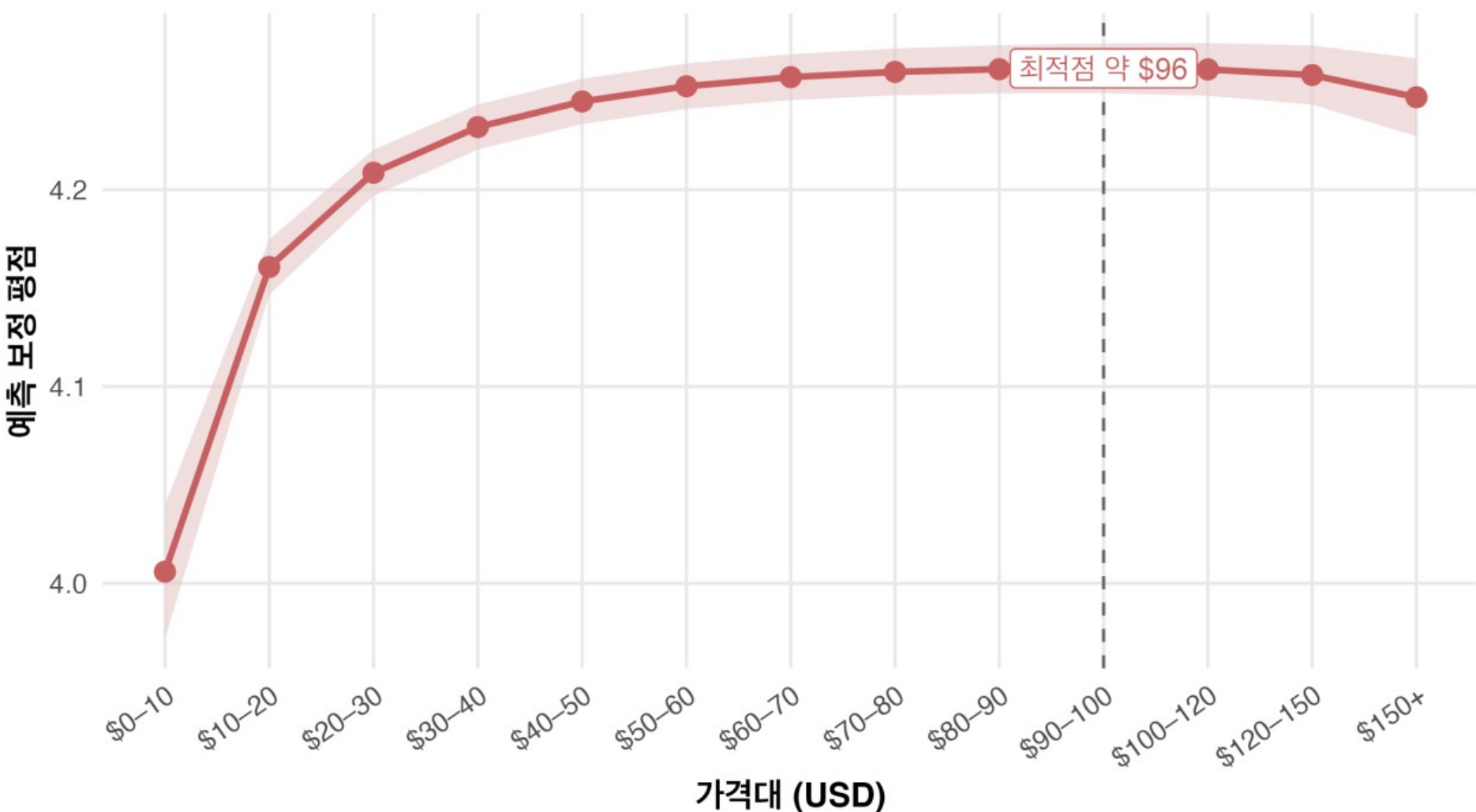
Products around \$96
show the highest predicted rating
- 

Model Fit

$R^2 = 0.074$ | $\text{Adj. } R^2 = 0.072$

가격과 보정 평점의 비선형 관계

가격대 구간별로 변환한 이차항 회귀 예측값



Price can work as a premium signal, but beyond a certain point, it becomes a risk

Premium strategy works **only within a consumer-acceptable price range**

Model Analysis 2: Exclusivity Moderation

H2 Result (DV: Adjusted Rating)

Conclusion: Exclusivity **strengthens price acceptance**

Regression Results and Interpretation

1 **Price** **$b_1 = 0.0446$** ($p < 0.05$)

Positive effect → Higher price, higher rating

2 **Exclusivity** **$b_2 = 0.033$** ($p < 0.05$)

Positive effect → Exclusive products have higher ratings

3 **Price × Exclusivity** **$b_3 = 0.0197$**

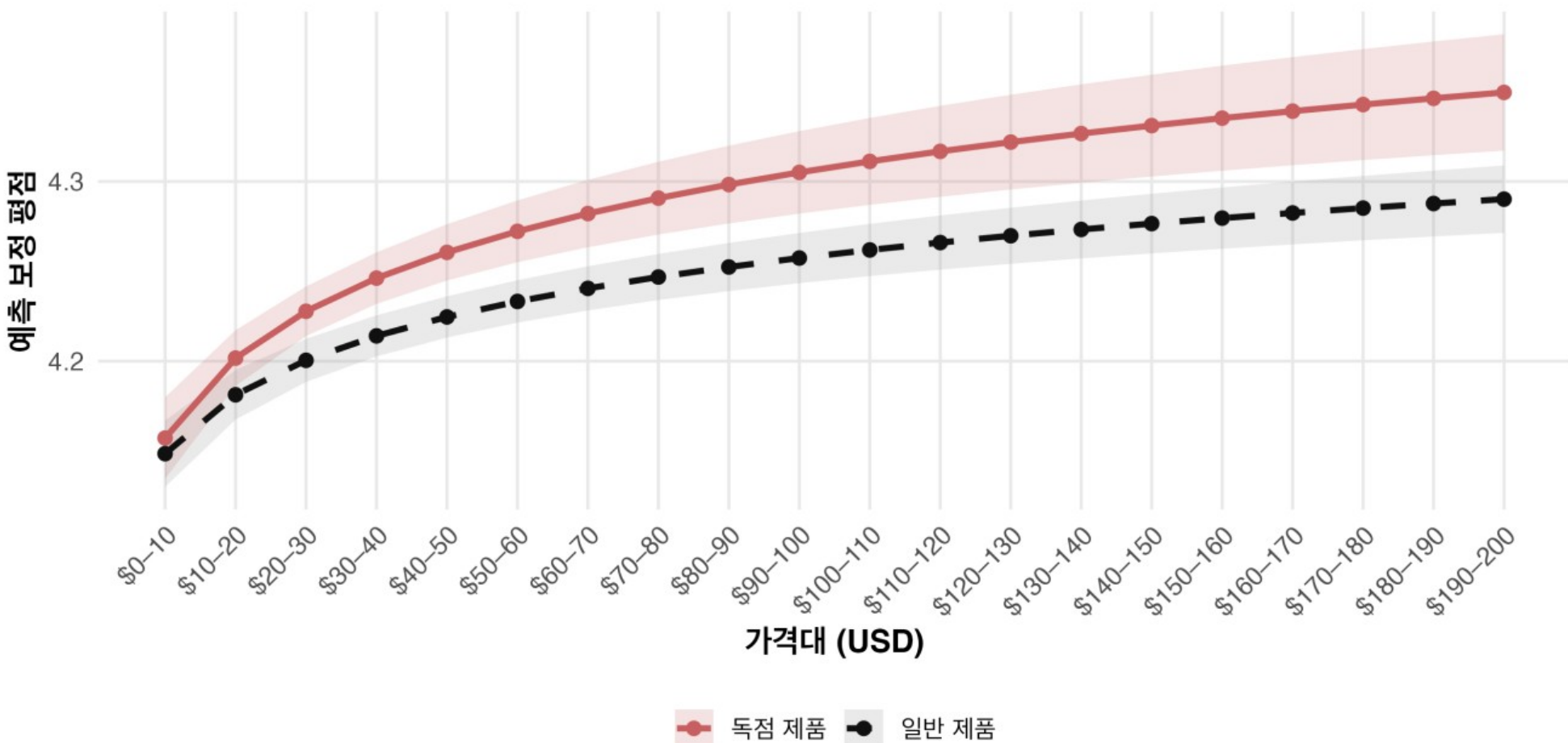
Positive interaction
→ Higher price is more acceptable ($p < 0.05$)

 **Model Fit**

$R^2 = 0.068$ | $\text{Adj. } R^2 = 0.066$

가격과 세포라 독점 여부의 상호작용

10달러 단위 가격 구간 기준 예측 보정 평점



Sephora exclusivity **strengthens the effect of premium pricing**

With the same price increase, exclusive products show a greater rating increase

Model Analysis 3: Vegan Claim Moderation

H3 Result (Adjusted Rating)

Regression Results and Interpretation

- 1

Price

$b1 = 0.0425$

$(p < 0.05)$

Positive effect → Higher price, higher rating
- 2

Vegan Claim

$b4 = 0.0114$

$(p > 0.05)$

Not significant → No clear main effect
- 3

Price × Vegan

$b5 = 0.0154$

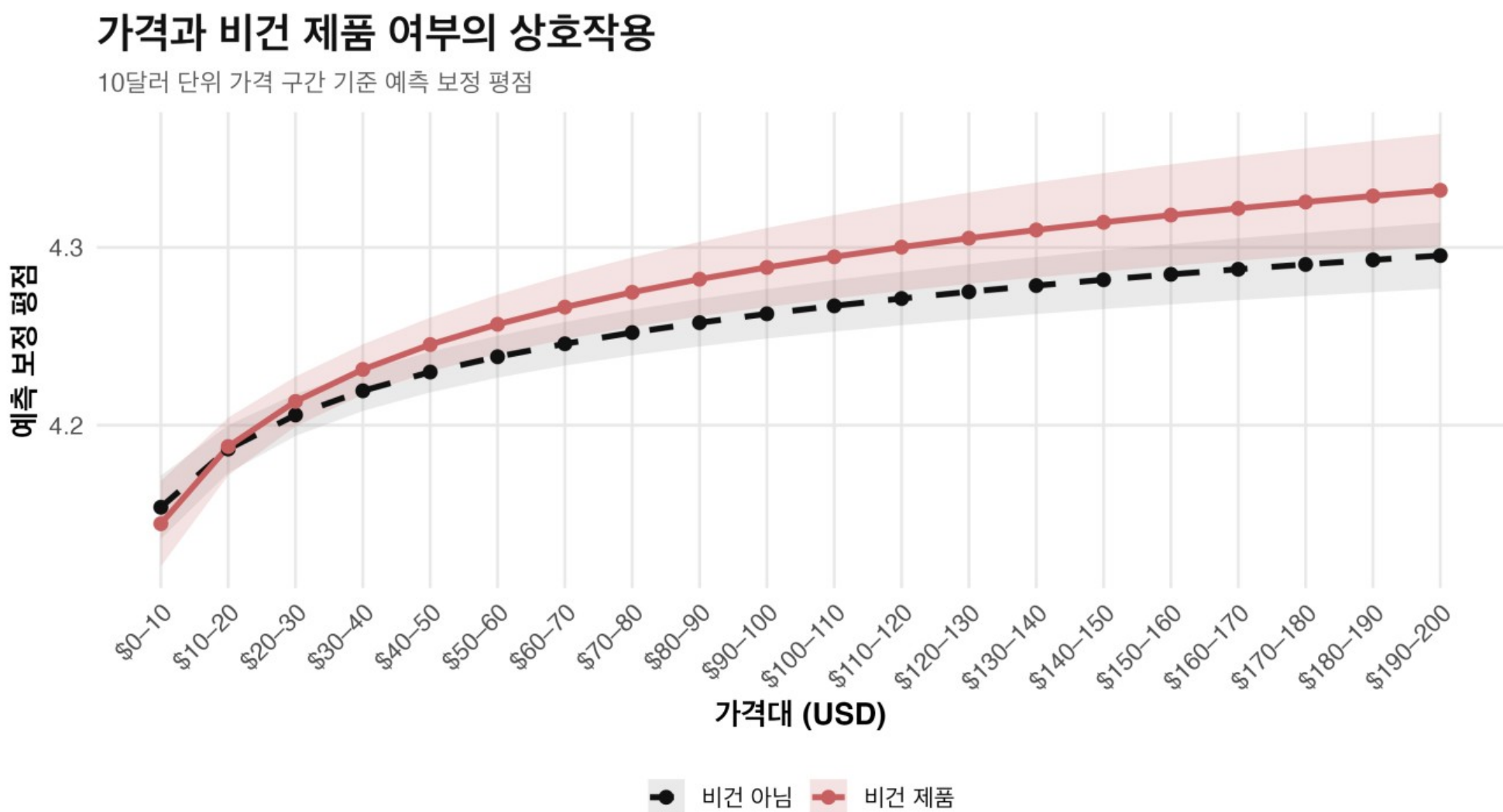
$(p > 0.05)$

Not significant
→ No clear moderation effect
- 

Model Fit

$R^2 = 0.069$ | $\text{Adj. } R^2 = 0.067$

Conclusion: No significant vegan moderation → Binary logit analysis added



The regression model shows **no clear vegan moderation effect**

Thus, **an additional binary logit model** tests the probability of high ratings

Additional Analysis 1: Binary Logit Analysis

Rechecking Vegan Claims with High-Rating Probability

Binary Logit Results and Interpretation

- 1

Price

$b = 0.281$ ($p < 0.05$)

Positive effect
→ Higher price increases high-rating probability
- 2

Vegan Claim

$b = 0.056$ ($p > 0.05$)

Not significant → Vegan status alone does not guarantee high ratings
- 3

Price × Vegan

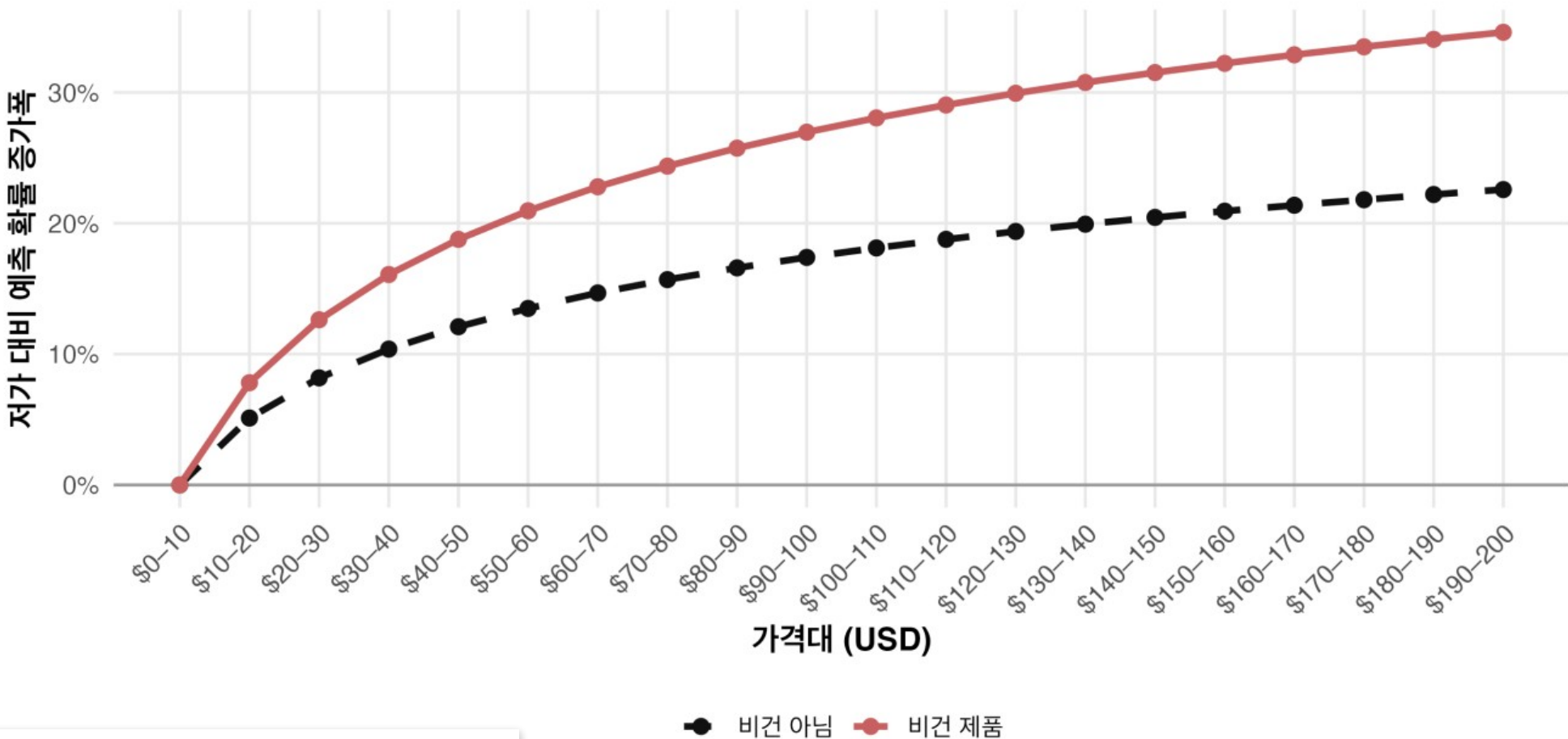
$b = 0.175$ ($p < 0.05$)

Positive interaction
→ Vegan claims work better at higher prices

Interpretation: Vegan moderation was unclear in average-rating regression, but **significant in the high-rating logit model**

가격 상승에 따른 높은 평가 확률 증가폭

각 집단의 가장 낮은 가격대 예측확률을 0으로 맞춘 비교



Vegan claims become more persuasive at higher price levels

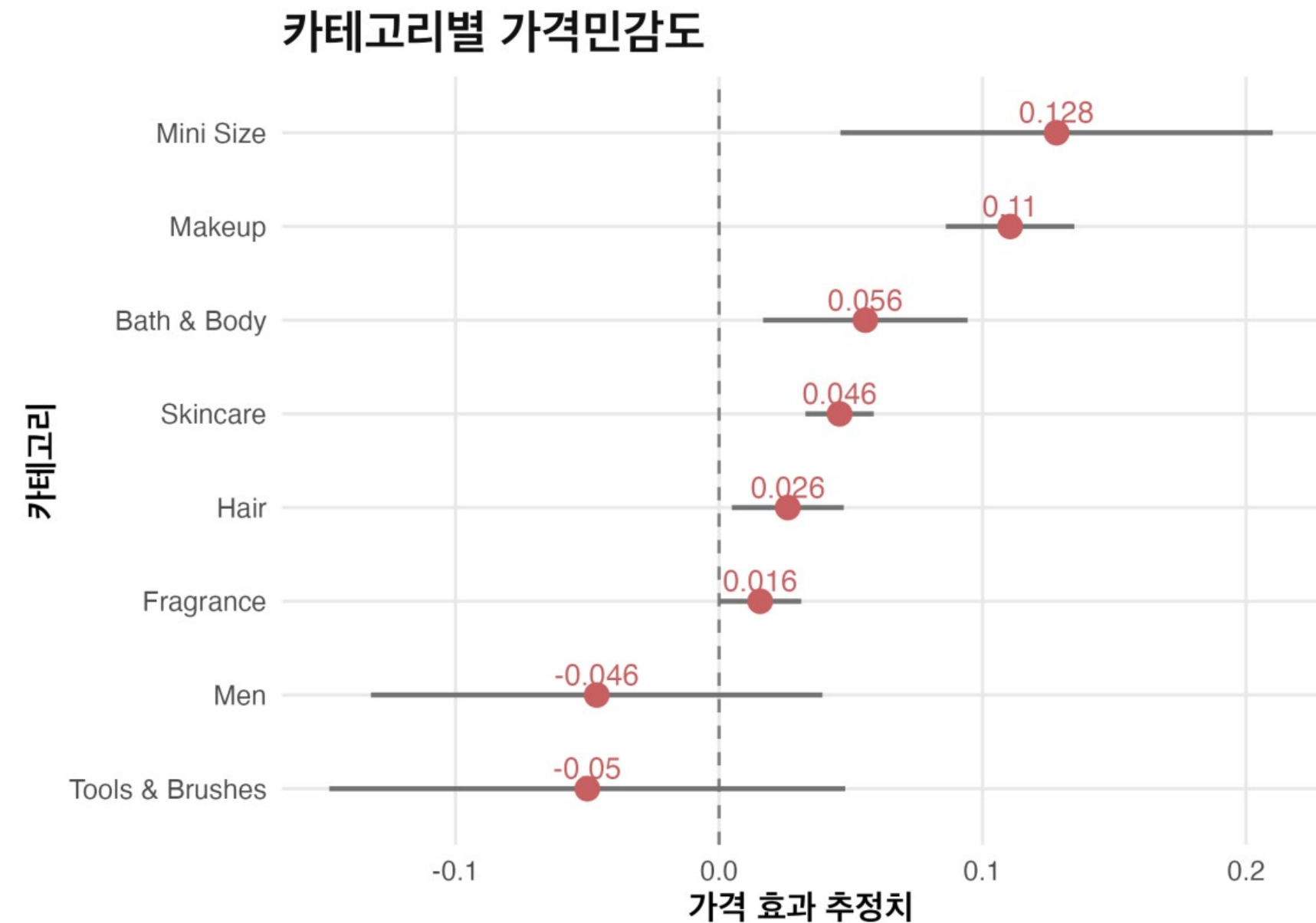
At low prices, they are weak value signals. At higher prices, they help justify premiums through ethics, ingredients, and brand philosophy

Additional Analysis 2: Category Effects

Comparing Price Effects by Category

Category Interpretation

- 1 Mini Size / Makeup**
 - Price works as a premium signal
 - Higher price raises quality expectations
- 2 Bath & Body / Skincare**
 - Price works as a quality signal
 - Higher price implies efficacy and ingredient value
- 3 Hair / Fragrance**
 - Price effect is limited
 - Usage, scent, and brand preference may matter more
- 4 Men / Tools & Brushes**
 - Price is less persuasive
 - Higher price does not improve evaluation



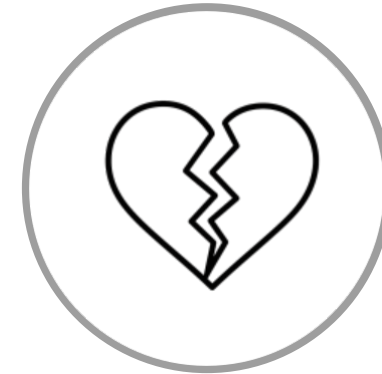
Price effects differ by category. Premium signals are strong in Mini Size and Makeup, but weak in Men and Tools & Brushes

Results and Discussion



Premium Signal

Price signals quality and brand value up to a certain level



Expectation Gap

Excessive prices can widen the gap between expectation and experience



Vegan Claim

Vegan claims justify premium prices better at higher price levels



Channel Trust

Exclusive channels signal trust and curation, not just higher price

“

Beauty satisfaction is not driven by price alone
Premium persuasion **depends on price range, exclusivity, vegan claims, and category context**

”

Managerial Implications

How can these findings guide pricing, distribution, and communication?

K-Beauty Brands

- Low prices alone do not guarantee satisfaction
- Brands **must justify price** through efficacy, quality, and storytelling

Distribution Channels

- Exclusive channels build **premium trust**
- Sales channels affect **price acceptance**

Pricing Strategy

- Higher is not always better. The **right price range** matters.
- Excessive prices can create **expectation mismatch** and lower satisfaction

Marketing Communication

- Vegan claims work better for **premium products**
- Align **value signals**:
vegan, scarcity, brand philosophy

Role-Based Implications



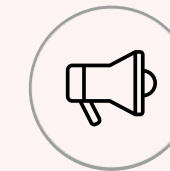
Brand Manager:

Positioning and launch pricing



MD/Distribution Manager:

Plan exclusive distribution and channel strategy



Marketer:

Create messages that justify the price



Strategy Team:

Identify categories suitable for premium strategy



Beauty pricing is not just setting a price tag. It means **creating a context consumers can accept**. **Premium pricing works** when price range, exclusivity, vegan claims, and category context align

Final Conclusion

“Beauty pricing is not just about setting a price tag
It is about designing reasons consumers can accept”



1. Price Principle

Higher is not always better
Price is most persuasive within an acceptable range

2. Reasons for Persuasion

Exclusivity and vegan claims work as signals that explain the price

3. Application Direction

Price acceptance differs by category. Premium strategies should be category-specific

THANK YOU